IT MASTERY
COURSE CATALOGUE

Networking Setup | Cloud Computing
Ethical Hacking | Penetration Testing

54 Courses - Full Module Contents - Tools Q&A - Learning Roadmap

Important Notice: Ethical Use Only

All hacking and penetration-testing content in this catalogue is for education and for authorised security work only. Only test systems you own or have explicit written permission to test. Unauthorised access is illegal under laws such as India's IT Act 2000, the US CFAA and the UK Computer Misuse Act. Practise in your own lab or in legal platforms (TryHackMe, Hack The Box, PortSwigger Academy, OWASP Juice Shop).

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A. NETWORKING SETUP

Course 1: Networking Fundamentals

Level: Beginner **Duration:** 4 weeks **Modules:** 8

- **Module 1:** What is a network, types (LAN/WAN/MAN/PAN)
- **Module 2:** Topologies: bus, star, ring, mesh, hybrid
- **Module 3:** OSI model - 7 layers in depth
- **Module 4:** TCP/IP model and encapsulation
- **Module 5:** Network devices: hub, switch, router, bridge, modem, AP
- **Module 6:** Cabling: UTP, fiber, coaxial, connectors
- **Module 7:** Basic troubleshooting: ping, tracert, ipconfig/ifconfig
- **Module 8:** Lab: build a small office network in Packet Tracer

Course 2: IP Addressing and Subnetting

Level: Beginner **Duration:** 3 weeks **Modules:** 7

- **Module 1:** Binary and decimal conversion
- **Module 2:** IPv4 classes A-E, private and public ranges
- **Module 3:** CIDR notation and subnet masks
- **Module 4:** Subnetting and VLSM
- **Module 5:** Supernetting and route summarisation
- **Module 6:** IPv6 addressing: types, EUI-64, compression
- **Module 7:** Lab: design an addressing plan for 5 branch offices

Course 3: CompTIA Network+ Prep

Level: Beginner **Duration:** 8 weeks **Modules:** 7

- **Module 1:** Networking concepts
- **Module 2:** Infrastructure and cabling
- **Module 3:** Network operations and monitoring
- **Module 4:** Network security basics
- **Module 5:** Troubleshooting methodology (7 steps)
- **Module 6:** Wireless standards 802.11a/b/g/n/ac/ax
- **Module 7:** Practice exams and exam strategy

Course 4: Cisco CCNA 200-301

Level: Intermediate **Duration:** 12 weeks **Modules:** 8

- **Module 1:** Network fundamentals
- **Module 2:** Network access: VLANs, trunking, STP, EtherChannel
- **Module 3:** IP connectivity: static routing, OSPF
- **Module 4:** IP services: NAT, DHCP, DNS, NTP, SNMP, QoS
- **Module 5:** Security fundamentals: ACLs, port security, AAA
- **Module 6:** Wireless LAN concepts and WLC
- **Module 7:** Automation and programmability: REST, JSON, Ansible
- **Module 8:** Full lab set and mock exams

Course 5: Cisco CCNP Enterprise (ENCOR)

Level: Advanced **Duration:** 12 weeks **Modules:** 8

- **Module 1:** Enterprise architecture and SD-Access
- **Module 2:** Dual-stack routing and virtualisation
- **Module 3:** EIGRP and OSPF advanced tuning
- **Module 4:** BGP fundamentals and path selection
- **Module 5:** Network assurance, telemetry, NetFlow
- **Module 6:** Security: 802.1X, TrustSec, CoPP
- **Module 7:** Automation: NETCONF, RESTCONF, YANG
- **Module 8:** Capstone enterprise design

Course 6: Routing Protocols Masterclass

Level: Intermediate **Duration:** 6 weeks **Modules:** 8

- **Module 1:** Static vs dynamic routing
- **Module 2:** RIP v1/v2
- **Module 3:** EIGRP metrics and feasibility condition
- **Module 4:** OSPF areas, LSAs, DR/BDR
- **Module 5:** IS-IS basics
- **Module 6:** BGP: eBGP, iBGP, attributes, route maps
- **Module 7:** Redistribution and route filtering
- **Module 8:** Lab: multi-protocol topology in GNS3

Course 7: Switching and VLAN Engineering

Level: Intermediate **Duration:** 5 weeks **Modules:** 7

- **Module 1:** Ethernet and MAC learning
- **Module 2:** VLANs and 802.1Q trunking
- **Module 3:** Inter-VLAN routing (router-on-a-stick, SVI)
- **Module 4:** Spanning Tree: STP, RSTP, MSTP
- **Module 5:** EtherChannel/LACP
- **Module 6:** Layer 3 switching
- **Module 7:** Switch security: DHCP snooping, DAI, port security

Course 8: Wireless Networking and Wi-Fi Design

Level: Intermediate **Duration:** 5 weeks **Modules:** 7

- **Module 1:** RF fundamentals: frequency, channels, interference
- **Module 2:** 802.11 standards and Wi-Fi 6/6E/7
- **Module 3:** Site survey and AP placement
- **Module 4:** WLAN controllers, SSIDs, roaming
- **Module 5:** WPA2/WPA3 and 802.1X enterprise security
- **Module 6:** Guest portals and captive networks
- **Module 7:** Troubleshooting Wi-Fi performance

Course 9: Network Security Fundamentals

Level: Intermediate **Duration:** 6 weeks **Modules:** 8

- **Module 1:** CIA triad and threat landscape
- **Module 2:** Firewalls: packet filter, stateful, NGFW
- **Module 3:** IDS/IPS concepts
- **Module 4:** VPN: IPsec, SSL/TLS, WireGuard
- **Module 5:** ACLs and segmentation
- **Module 6:** NAC and 802.1X
- **Module 7:** Zero Trust principles
- **Module 8:** Security policy and hardening checklists

Course 10: Firewall Administration (Fortinet / Palo Alto / pfSense)

Level: Intermediate **Duration:** 6 weeks **Modules:** 7

- **Module 1:** Firewall architecture and zones
- **Module 2:** Policies, NAT and VIPs
- **Module 3:** Application control and URL filtering
- **Module 4:** Site-to-site and remote-access VPN
- **Module 5:** High availability
- **Module 6:** Logging and SIEM integration
- **Module 7:** Lab: deploy pfSense with VLANs

Course 11: Linux Networking

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

- **Module 1:** ip, ss, nmcli, netplan
- **Module 2:** Bridges, bonding, VLAN interfaces
- **Module 3:** iptables and nftables
- **Module 4:** DNS (BIND/unbound) and DHCP servers
- **Module 5:** Routing and IP forwarding
- **Module 6:** tcpdump and traffic analysis
- **Module 7:** Network namespaces and containers

Course 12: Network Automation with Python and Ansible

Level: Advanced **Duration:** 6 weeks **Modules:** 7

- **Module 1:** Python for network engineers
- **Module 2:** Netmiko, NAPALM, Nornir
- **Module 3:** Jinja2 templates
- **Module 4:** Ansible network modules and playbooks
- **Module 5:** REST APIs, JSON, YAML
- **Module 6:** Git and CI/CD for network configs
- **Module 7:** Lab: automate VLAN rollout on 20 switches

Course 13: SD-WAN and VPN Technologies

Level: Advanced **Duration:** 4 weeks **Modules:** 7

- **Module 1:** WAN technologies: MPLS, leased lines, broadband
- **Module 2:** DMVPN and GRE
- **Module 3:** IPsec in depth
- **Module 4:** SD-WAN architecture (controller, edge, orchestrator)
- **Module 5:** Application-aware routing
- **Module 6:** Cloud on-ramp
- **Module 7:** Monitoring and SLAs

Course 14: Network Monitoring and Troubleshooting

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

- **Module 1:** Baselines and KPIs
- **Module 2:** SNMP, syslog, NetFlow/sFlow
- **Module 3:** Zabbix, Nagios, PRTG, LibreNMS
- **Module 4:** Wireshark analysis
- **Module 5:** Latency, jitter, packet loss diagnosis
- **Module 6:** Layered troubleshooting approach
- **Module 7:** Incident documentation

Course 15: Network Design and Architecture

Level: Advanced **Duration:** 6 weeks **Modules:** 7

- **Module 1:** Hierarchical design: core, distribution, access
- **Module 2:** Spine-leaf and data-centre fabrics
- **Module 3:** High availability and redundancy
- **Module 4:** Capacity planning
- **Module 5:** QoS design for voice and video
- **Module 6:** IPv6 migration strategy
- **Module 7:** Documentation and diagram standards

B. CLOUD COMPUTING

Course 16: Cloud Computing Fundamentals

Level: Beginner **Duration:** 3 weeks **Modules:** 7

- **Module 1:** Cloud concepts and history
- **Module 2:** IaaS, PaaS, SaaS, FaaS
- **Module 3:** Public, private, hybrid, multi-cloud
- **Module 4:** Shared responsibility model
- **Module 5:** Virtualisation and containers overview
- **Module 6:** Cost models and TCO
- **Module 7:** Cloud providers comparison

Course 17: AWS Cloud Practitioner

Level: Beginner **Duration:** 4 weeks **Modules:** 7

- **Module 1:** AWS global infrastructure
- **Module 2:** IAM basics
- **Module 3:** EC2, S3, EBS, VPC overview
- **Module 4:** Pricing, billing, support plans
- **Module 5:** Well-Architected Framework
- **Module 6:** Security and compliance
- **Module 7:** Exam preparation

Course 18: AWS Solutions Architect Associate

Level: Intermediate **Duration:** 10 weeks **Modules:** 10

- **Module 1:** IAM, Organizations, SCPs
- **Module 2:** VPC: subnets, route tables, NAT, endpoints, peering, Transit Gateway
- **Module 3:** EC2, Auto Scaling, ELB
- **Module 4:** S3, EFS, FSx, storage classes
- **Module 5:** RDS, Aurora, DynamoDB
- **Module 6:** Route 53, CloudFront
- **Module 7:** SQS, SNS, EventBridge
- **Module 8:** Serverless: Lambda, API Gateway
- **Module 9:** Monitoring: CloudWatch, CloudTrail
- **Module 10:** Mock exams

Course 19: AWS SysOps and DevOps Engineer

Level: Advanced **Duration:** 8 weeks **Modules:** 7

- **Module 1:** CloudFormation and CDK
- **Module 2:** CodePipeline, CodeBuild, CodeDeploy
- **Module 3:** Systems Manager
- **Module 4:** Config, Inspector, GuardDuty
- **Module 5:** Blue/green and canary deployments
- **Module 6:** Disaster recovery strategies
- **Module 7:** Cost optimisation

Course 20: Microsoft Azure Fundamentals (AZ-900)

Level: Beginner **Duration:** 3 weeks **Modules:** 6

- **Module 1:** Azure architecture: regions, subscriptions, resource groups
- **Module 2:** Compute, networking, storage services
- **Module 3:** Identity: Entra ID, RBAC
- **Module 4:** Governance: Policy, Blueprints, Cost Management
- **Module 5:** Security: Defender for Cloud
- **Module 6:** Monitoring: Azure Monitor

Course 21: Azure Administrator (AZ-104)

Level: Intermediate **Duration:** 8 weeks **Modules:** 7

- **Module 1:** Entra ID users, groups, RBAC
- **Module 2:** VMs, scale sets, App Service, containers
- **Module 3:** Virtual networks, NSGs, load balancers, VPN Gateway
- **Module 4:** Storage accounts, blobs, files
- **Module 5:** Backup and Site Recovery
- **Module 6:** Azure Monitor and Log Analytics
- **Module 7:** ARM templates and Bicep

Course 22: Azure Solutions Architect (AZ-305)

Level: Advanced **Duration:** 8 weeks **Modules:** 7

- **Module 1:** Identity and governance design
- **Module 2:** Data storage design
- **Module 3:** Business continuity design
- **Module 4:** Infrastructure and network design
- **Module 5:** Hybrid connectivity: ExpressRoute
- **Module 6:** Migration strategies
- **Module 7:** Landing zones

Course 23: Google Cloud Associate Engineer

Level: Intermediate **Duration:** 8 weeks **Modules:** 7

- **Module 1:** Projects, IAM, resource hierarchy
- **Module 2:** Compute Engine, GKE, Cloud Run
- **Module 3:** VPC, firewall, load balancing
- **Module 4:** Cloud Storage, Cloud SQL, BigQuery
- **Module 5:** Cloud Monitoring and Logging
- **Module 6:** Deployment Manager and Terraform
- **Module 7:** Billing and quotas

Course 24: Docker and Containers

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

- **Module 1:** Container concepts vs VMs
- **Module 2:** Images, Dockerfiles, layers
- **Module 3:** Volumes and networking
- **Module 4:** Docker Compose
- **Module 5:** Registries and image scanning
- **Module 6:** Container security best practices
- **Module 7:** Lab: containerise a 3-tier app

Course 25: Kubernetes Administration (CKA)

Level: Advanced **Duration:** 8 weeks **Modules:** 8

- **Module 1:** Cluster architecture
- **Module 2:** Pods, Deployments, StatefulSets, DaemonSets
- **Module 3:** Services, Ingress, NetworkPolicy
- **Module 4:** ConfigMaps, Secrets, RBAC
- **Module 5:** Storage: PV, PVC, StorageClass
- **Module 6:** Scheduling, taints, affinity
- **Module 7:** Cluster install with kubeadm, upgrades, etcd backup
- **Module 8:** Troubleshooting

Course 26: Terraform and Infrastructure as Code

Level: Intermediate **Duration:** 5 weeks **Modules:** 7

- **Module 1:** IaC principles
- **Module 2:** HCL syntax, providers, resources
- **Module 3:** State management and remote backends
- **Module 4:** Modules and workspaces
- **Module 5:** Multi-cloud deployments
- **Module 6:** Policy as code (Sentinel, OPA)
- **Module 7:** CI/CD integration

Course 27: DevOps and CI/CD Pipelines

Level: Intermediate **Duration:** 6 weeks **Modules:** 7

- **Module 1:** DevOps culture and lifecycle
- **Module 2:** Git and branching strategies
- **Module 3:** Jenkins, GitHub Actions, GitLab CI
- **Module 4:** Build, test, deploy automation
- **Module 5:** Artifact repositories
- **Module 6:** Monitoring: Prometheus and Grafana
- **Module 7:** DevSecOps integration

Course 28: Cloud Networking and Hybrid Connectivity

Level: Advanced **Duration:** 5 weeks **Modules:** 7

- **Module 1:** Cloud VPC/VNet design
- **Module 2:** Peering, Transit Gateway, hub-and-spoke
- **Module 3:** VPN and Direct Connect/ExpressRoute
- **Module 4:** DNS in cloud, private endpoints
- **Module 5:** Load balancers and WAF
- **Module 6:** Network security groups and firewalls
- **Module 7:** Multi-region architecture

Course 29: Cloud Security (CCSP / Security+ Cloud)

Level: Advanced **Duration:** 8 weeks **Modules:** 7

- **Module 1:** Cloud security architecture
- **Module 2:** Identity federation, SSO, MFA
- **Module 3:** Data encryption, KMS, HSM
- **Module 4:** Logging and SIEM in cloud
- **Module 5:** CSPM and misconfiguration detection
- **Module 6:** Compliance: ISO 27001, SOC 2, GDPR, HIPAA
- **Module 7:** Incident response in cloud

Course 30: Serverless and Cloud Migration

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

- **Module 1:** Serverless architecture patterns
- **Module 2:** Lambda, Azure Functions, Cloud Functions
- **Module 3:** Event-driven design
- **Module 4:** The 6 Rs of migration
- **Module 5:** Assessment and discovery tools
- **Module 6:** Database migration
- **Module 7:** Post-migration optimisation

C. ETHICAL HACKING

Course 31: Introduction to Ethical Hacking

Level: Beginner **Duration:** 4 weeks **Modules:** 6

- **Module 1:** Hacker types: white, grey, black hat
- **Module 2:** Cyber kill chain and MITRE ATT&CK
- **Module 3:** Legal, ethics and laws (IT Act, CFAA, GDPR)
- **Module 4:** Setting up a home lab: VirtualBox, Kali, Metasploitable
- **Module 5:** Basic Linux for hackers
- **Module 6:** Reporting and responsible disclosure

Course 32: CEH v13 Certified Ethical Hacker Prep

Level: Intermediate **Duration:** 12 weeks **Modules:** 10

- **Module 1:** Footprinting and reconnaissance
- **Module 2:** Scanning networks
- **Module 3:** Enumeration
- **Module 4:** Vulnerability analysis
- **Module 5:** System hacking concepts
- **Module 6:** Malware threats (concepts and defence)
- **Module 7:** Sniffing, social engineering, DoS
- **Module 8:** Session hijacking, IDS/firewall evasion concepts
- **Module 9:** Web server and web application hacking
- **Module 10:** SQL injection, wireless, mobile, IoT, cloud, cryptography

Course 33: Linux for Hackers

Level: Beginner **Duration:** 3 weeks **Modules:** 7

- **Module 1:** File system and permissions
- **Module 2:** Bash scripting
- **Module 3:** Process and service management
- **Module 4:** Networking commands
- **Module 5:** Package management
- **Module 6:** Log analysis
- **Module 7:** Privilege model: sudo, SUID, capabilities

Course 34: Reconnaissance and OSINT

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

- **Module 1:** Passive vs active recon
- **Module 2:** WHOIS, DNS, subdomain discovery
- **Module 3:** Search-engine dorking
- **Module 4:** Shodan and Censys
- **Module 5:** theHarvester, Maltego, Recon-ng
- **Module 6:** Social media OSINT ethics
- **Module 7:** Documenting findings

Course 35: Network Scanning and Enumeration

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

- **Module 1:** TCP/UDP scanning theory
- **Module 2:** Nmap scan types and timing
- **Module 3:** NSE scripts
- **Module 4:** Service and OS fingerprinting
- **Module 5:** SMB, SNMP, LDAP, DNS enumeration
- **Module 6:** Vulnerability scanners: Nessus, OpenVAS
- **Module 7:** Reading and triaging scan results

Course 36: Web Application Hacking (OWASP Top 10)

Level: Intermediate **Duration:** 8 weeks **Modules:** 10

- **Module 1:** HTTP, cookies, sessions
- **Module 2:** Broken access control
- **Module 3:** Injection: SQL, command, LDAP
- **Module 4:** XSS: reflected, stored, DOM
- **Module 5:** CSRF and SSRF
- **Module 6:** Authentication and session flaws
- **Module 7:** Insecure deserialisation, XXE
- **Module 8:** Security misconfiguration
- **Module 9:** Burp Suite workflow
- **Module 10:** Labs: DVWA, Juice Shop, WebGoat

Course 37: Wireless Network Hacking and Defence

Level: Intermediate **Duration:** 4 weeks **Modules:** 6

- **Module 1:** WLAN security protocols WEP/WPA/WPA2/WPA3
- **Module 2:** Monitor mode and capture
- **Module 3:** Handshake analysis in lab environments
- **Module 4:** Rogue AP and evil twin detection
- **Module 5:** Wireless IDS
- **Module 6:** Hardening enterprise Wi-Fi

Course 38: Social Engineering and Phishing Awareness

Level: Beginner **Duration:** 3 weeks **Modules:** 6

- **Module 1:** Psychology of manipulation
- **Module 2:** Phishing, vishing, smishing, pretexting
- **Module 3:** Physical security and tailgating
- **Module 4:** Running authorised phishing simulations (GoPhish)
- **Module 5:** Awareness training design
- **Module 6:** Metrics and reporting

Course 39: Cryptography for Security Professionals

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

- **Module 1:** Symmetric vs asymmetric encryption
- **Module 2:** Hashing and digital signatures
- **Module 3:** PKI and certificates
- **Module 4:** TLS handshake
- **Module 5:** Common crypto failures
- **Module 6:** Password storage and cracking theory
- **Module 7:** Tools: OpenSSL, GPG, Hashcat (authorised)

Course 40: Mobile and IoT Security

Level: Advanced **Duration:** 5 weeks **Modules:** 7

- **Module 1:** Android and iOS architecture
- **Module 2:** OWASP Mobile Top 10
- **Module 3:** Static and dynamic app analysis
- **Module 4:** MobSF, Frida basics
- **Module 5:** IoT protocols: MQTT, Zigbee, BLE
- **Module 6:** Firmware analysis concepts
- **Module 7:** Securing devices

Course 41: Malware Analysis Fundamentals

Level: Advanced **Duration:** 6 weeks **Modules:** 7

- **Module 1:** Malware families and behaviour
- **Module 2:** Safe analysis lab setup
- **Module 3:** Static analysis: strings, PE headers
- **Module 4:** Dynamic analysis and sandboxes
- **Module 5:** Basic reverse engineering with Ghidra
- **Module 6:** YARA rules
- **Module 7:** Writing analysis reports

Course 42: Digital Forensics and Incident Response

Level: Advanced **Duration:** 6 weeks **Modules:** 7

- **Module 1:** Forensic principles and chain of custody
- **Module 2:** Disk and memory acquisition
- **Module 3:** Autopsy, FTK Imager, Volatility
- **Module 4:** Windows and Linux artefacts
- **Module 5:** Log and timeline analysis
- **Module 6:** IR lifecycle: NIST 800-61
- **Module 7:** Reporting for legal use

D. PENETRATION TESTING

Course 43: Penetration Testing Methodology

Level: Intermediate **Duration:** 4 weeks **Modules:** 6

- **Module 1:** PTES, OSSTMM, NIST 800-115
- **Module 2:** Scoping, rules of engagement, contracts
- **Module 3:** Threat modelling
- **Module 4:** Phases: recon, exploit, post-exploit, report
- **Module 5:** Risk rating with CVSS
- **Module 6:** Client communication

Course 44: eJPT / Junior Penetration Tester

Level: Beginner **Duration:** 6 weeks **Modules:** 7

- **Module 1:** Networking and web basics refresher
- **Module 2:** Host discovery and port scanning
- **Module 3:** Vulnerability assessment
- **Module 4:** Metasploit framework basics
- **Module 5:** Password attacks in lab
- **Module 6:** Pivoting basics
- **Module 7:** Exam-style practice labs

Course 45: CompTIA PenTest+

Level: Intermediate **Duration:** 8 weeks **Modules:** 6

- **Module 1:** Planning and scoping
- **Module 2:** Information gathering and vulnerability scanning
- **Module 3:** Attacks and exploits
- **Module 4:** Reporting and communication
- **Module 5:** Tools and code analysis
- **Module 6:** Remediation recommendations

Course 46: OSCP-Style Penetration Testing

Level: Advanced **Duration:** 16 weeks **Modules:** 9

- **Module 1:** Enumeration mastery
- **Module 2:** Linux privilege escalation
- **Module 3:** Windows privilege escalation
- **Module 4:** Active Directory attacks
- **Module 5:** Buffer overflow concepts
- **Module 6:** Web exploitation chains
- **Module 7:** Tunnelling and pivoting
- **Module 8:** Proof of concept and report writing
- **Module 9:** Hack The Box and TryHackMe practice

Course 47: Active Directory Penetration Testing

Level: Advanced **Duration:** 6 weeks **Modules:** 7

- **Module 1:** AD architecture: forests, domains, GPO
- **Module 2:** Enumeration: BloodHound, LDAP
- **Module 3:** Kerberos attacks (Kerberoasting, AS-REP)
- **Module 4:** Credential abuse and lateral movement
- **Module 5:** Delegation and ACL abuse
- **Module 6:** Persistence concepts
- **Module 7:** Detection and hardening

Course 48: Web Application Penetration Testing (BSCP-Style)

Level: Advanced **Duration:** 8 weeks **Modules:** 7

- **Module 1:** Advanced authentication bypass
- **Module 2:** Business logic flaws
- **Module 3:** API security testing
- **Module 4:** JWT and OAuth weaknesses
- **Module 5:** Race conditions and cache poisoning
- **Module 6:** Advanced SQLi and XSS
- **Module 7:** Burp extensions and automation

Course 49: Network Penetration Testing

Level: Intermediate **Duration:** 6 weeks **Modules:** 7

- **Module 1:** External and internal testing
- **Module 2:** Service exploitation with Metasploit
- **Module 3:** Password attacks and spraying
- **Module 4:** MITM and ARP concepts in lab
- **Module 5:** VLAN hopping and segmentation testing
- **Module 6:** Post-exploitation and data discovery
- **Module 7:** Cleanup

Course 50: Cloud Penetration Testing (AWS/Azure/GCP)

Level: Advanced **Duration:** 6 weeks **Modules:** 7

- **Module 1:** Cloud attack surface
- **Module 2:** IAM misconfiguration abuse
- **Module 3:** S3 and storage exposure
- **Module 4:** Metadata service risks
- **Module 5:** Container and Kubernetes testing
- **Module 6:** Tools: ScoutSuite, Pacu, Prowler
- **Module 7:** Cloud provider testing policies

Course 51: Red Team Operations

Level: Advanced **Duration:** 8 weeks **Modules:** 7

- **Module 1:** Red vs purple vs blue teaming
- **Module 2:** Adversary emulation with ATT&CK
- **Module 3:** C2 framework concepts
- **Module 4:** Initial access and OPSEC principles
- **Module 5:** Lateral movement and persistence
- **Module 6:** Detection engineering feedback
- **Module 7:** Operation reports

Course 52: Vulnerability Management and Reporting

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

- **Module 1:** Asset inventory
- **Module 2:** Scanning cadence and tools
- **Module 3:** CVSS, EPSS, KEV prioritisation
- **Module 4:** Patch management
- **Module 5:** Writing executive and technical reports
- **Module 6:** Remediation tracking
- **Module 7:** Metrics and dashboards

Course 53: Bug Bounty Hunting

Level: Intermediate **Duration:** 6 weeks **Modules:** 7

- **Module 1:** Platforms: HackerOne, Bugcrowd
- **Module 2:** Scope reading and legal safe harbour
- **Module 3:** Recon automation
- **Module 4:** Common high-impact bug classes
- **Module 5:** Writing quality reports
- **Module 6:** Triage and communication
- **Module 7:** Ethics and responsible disclosure

Course 54: Blue Team and SOC Analyst

Level: Intermediate **Duration:** 8 weeks **Modules:** 7

- **Module 1:** SOC structure and tiers
- **Module 2:** SIEM: Splunk, Elastic, Wazuh
- **Module 3:** Log sources and correlation
- **Module 4:** Threat hunting
- **Module 5:** Sigma and detection rules
- **Module 6:** Incident handling playbooks
- **Module 7:** Threat intelligence

E. TOOLS - QUESTION AND ANSWER (FULL DETAILS)

Tool 1: Wireshark

Category: Network Analysis

Overview: Open-source packet analyser that captures and dissects traffic on a network interface in real time.

Usage / Examples:

```
Capture filter: port 80 | Display filters: ip.addr==192.168.1.10, tcp.flags.syn==1,
http.request, dns
```

Q: What is Wireshark used for?

A: Troubleshooting network issues, analysing protocols, finding latency or retransmissions, and investigating suspicious traffic.

Q: Capture filter vs display filter?

A: Capture filters (BPF syntax) limit what is recorded; display filters hide packets already captured and are more flexible.

Q: Can it decrypt HTTPS?

A: Only if you supply the TLS session keys (SSLKEYLOGFILE) or the server private key for non-forward-secret suites.

Tool 2: Nmap

Category: Scanning

Overview: Network mapper for host discovery, port scanning, service/version detection and OS fingerprinting.

Usage / Examples:

```
nmap -sn 192.168.1.0/24 | nmap -sV -p 1-1000 target | nmap -A target | nmap --script vuln
target
```

Q: What does -sV do?

A: Probes open ports to determine service name and version.

Q: Difference between -sS and -sT?

A: -sS is a half-open SYN scan needing raw-socket privileges; -sT completes the full TCP handshake using the OS connect() call.

Q: What is NSE?

A: The Nmap Scripting Engine runs Lua scripts for discovery, vulnerability checks and brute-force audits, only on systems you are authorised to test.

Tool 3: Kali Linux

Category: Platform

Overview: Debian-based distribution maintained by Offensive Security that bundles hundreds of security tools.

Usage / Examples:

```
Install as VM or WSL; update with: sudo apt update && sudo apt full-upgrade
```

Q: Is Kali good for beginners?

A: It is for security learning, but not a general desktop OS; learn Linux basics first.

Q: Why run it in a VM?

A: Isolation, snapshots, and safety for lab experiments.

Q: Alternatives?

A: Parrot OS, BlackArch, or a custom toolset on Ubuntu.

Tool 4: Metasploit Framework

Category: Exploitation Framework

Overview: Modular framework for developing, testing and executing exploit modules against authorised targets.

Usage / Examples:

```
msfconsole | search type:exploit name:smb | use <module> | show options | set RHOSTS <ip> | run
```

Q: What are modules?

A: Exploits, auxiliary scanners, payloads, encoders, post-exploitation and evasion modules.

Q: What is Meterpreter?

A: An advanced in-memory payload giving a flexible post-exploitation session.

Q: Is it legal?

A: Yes when used with written permission on systems you own or are contracted to test.

Tool 5: Burp Suite

Category: Web Testing

Overview: Intercepting proxy and toolkit for testing web application security.

Usage / Examples:

```
Proxy, Repeater, Intruder, Scanner (Pro), Decoder, Comparer, Collaborator
```

Q: How do I start?

A: Configure the browser proxy to 127.0.0.1:8080 and install Burp's CA certificate.

Q: Repeater vs Intruder?

A: Repeater manually resends one modified request; Intruder automates many payload variations.

Q: Community vs Professional?

A: Pro adds the automated scanner, saved projects and no throttling of Intruder.

Tool 6: OWASP ZAP

Category: Web Testing

Overview: Free open-source web app scanner and intercepting proxy from OWASP.

Usage / Examples:

```
zap-baseline.py -t https://test.example | Active scan, spider, AJAX spider, API scan
```

Q: ZAP vs Burp?

A: ZAP is free and automation-friendly; Burp has a more polished manual workflow.

Q: Passive vs active scan?

A: Passive only observes traffic; active sends attack payloads and must be authorised.

Q: Use in CI/CD?

A: Yes, via Docker images and baseline scan scripts.

Tool 7: Nessus / OpenVAS

Category: Vulnerability Scanning

Overview: Scanners that check hosts for known vulnerabilities, misconfigurations and missing patches.

Usage / Examples:

```
Create scan template, set targets and credentials, schedule, export report
```

Q: Credentialed vs non-credentialed?

A: Credentialed scans log in and give deeper, more accurate patch data.

Q: False positives?

A: Verify manually and tune plugins; never report unverified findings as fact.

Q: OpenVAS?

A: The open-source engine within Greenbone Vulnerability Management.

Tool 8: John the Ripper and Hashcat

Category: Password Auditing

Overview: Offline password-hash auditing tools used to test password strength in authorised assessments.

Usage / Examples:

```
john --wordlist=list.txt hashes.txt | hashcat -m 0 -a 0 hashes.txt list.txt
```

Q: Why audit password hashes?
A: To prove weak passwords exist and drive policy such as length, MFA and breached-password checks.

Q: CPU vs GPU?
A: Hashcat uses GPUs for far higher speed on fast hashes.

Q: How to defend?
A: Use slow salted hashes (bcrypt, scrypt, Argon2), long passphrases and MFA.

Tool 9: Hydra

Category: Credential Testing

Overview: Online login tester supporting many protocols, used to verify account-lockout and weak-credential exposure.

Usage / Examples:

```
hydra -L users.txt -P pass.txt ssh://target (authorised lab only)
```

Q: Risk of use?
A: Can lock accounts and trigger alerts; always coordinate with the system owner.

Q: Defences?
A: Lockout thresholds, rate limiting, MFA, fail2ban.

Q: Alternatives?
A: Medusa, Ncrack, CrackMapExec for AD contexts.

Tool 10: Aircrack-ng Suite

Category: Wireless

Overview: Set of tools for auditing Wi-Fi security on networks you own.

Usage / Examples:

```
airmon-ng, airodump-ng, aireplay-ng, aircrack-ng
```

Q: What does it assess?
A: Strength of WEP/WPA/WPA2 pre-shared keys and wireless configuration.

Q: Why monitor mode?
A: It lets the adapter capture frames not addressed to it.

Q: Defence?
A: Use WPA3 or WPA2-Enterprise with long, random keys.

Tool 11: Netcat

Category: Networking Utility

Overview: The 'Swiss-army knife' for reading and writing data over TCP/UDP.

Usage / Examples:

```
nc -lvnp 4444 | nc host 80 | nc -zv host 20-100
```

Q: Common legit uses?
A: Port checks, file transfers, banner grabbing, debugging services.

Q: Ncat?
A: Nmap's modern rewrite with SSL and access-control options.

Q: Is it dangerous?
A: It is neutral, but its misuse is flagged by security tools.

Tool 12: tcpdump

Category: Network Analysis

Overview: Command-line packet capture tool.

Usage / Examples:

```
tcpdump -i eth0 port 53 | tcpdump -nn -w out.pcap host 10.0.0.5
```

Q: -n flag?
A: Disables name resolution for faster, cleaner output.

Q: Read a file?
A: tcpdump -r out.pcap.

Q: Why use it over Wireshark?
A: Works on headless servers over SSH.

Tool 13: Cisco Packet Tracer

Category: Simulation

Overview: Free network simulator for building and testing Cisco topologies.

Usage / Examples:

```
Drag devices, connect cables, configure IOS CLI, simulation mode
```

Q: Who is it for?
A: CCNA students and beginners.

Q: Limits?
A: Not all IOS features are supported; use GNS3/EVE-NG for real images.

Q: Where to get it?
A: Free via Cisco Networking Academy.

Tool 14: GNS3 / EVE-NG

Category: Simulation

Overview: Emulators that run real router, switch and firewall images in virtual labs.

Usage / Examples:

```
Import images, build topology, connect to VMs
```

Q: Difference from Packet Tracer?
A: They emulate actual OS images, giving realistic behaviour.

Q: Hardware needs?
A: 16 GB+ RAM recommended for larger labs.

Q: Licensing?
A: You must supply legitimately licensed vendor images.

Tool 15: SolarWinds / Zabbix / PRTG

Category: Monitoring

Overview: Network monitoring platforms for availability, performance and alerting.

Usage / Examples:

```
SNMP polling, traps, dashboards, thresholds
```

Q: What is SNMP?
A: A protocol to query and manage devices; v3 adds authentication and encryption.

Q: Agent vs agentless?
A: Agents give deeper host metrics; agentless uses SNMP, WMI or ICMP.

Q: Free option?
A: Zabbix and LibreNMS are open source.

Tool 16: pfSense / OPNsense

Category: Firewall

Overview: Open-source firewall/router distributions.

Usage / Examples:

```
Interfaces, rules, NAT, VPN (OpenVPN, WireGuard, IPsec), Suricata/Snort packages
```

Q: Why use in labs?
A: Realistic segmentation and VPN practice at no cost.

Q: Rule order?
A: Rules are evaluated top-down; first match wins.

Q: HA?
A: CARP and pfsync provide failover.

Tool 17: Snort / Suricata

Category: IDS/IPS

Overview: Network intrusion detection and prevention engines using signature rules.

Usage / Examples:

```
alert tcp any any -> any 80 (msg:'test'; sid:1000001;)
```

Q: IDS vs IPS?
A: IDS alerts; IPS can block inline.

Q: Suricata advantage?
A: Multi-threading, TLS logging, EVE JSON output.

Q: Rules source?
A: Emerging Threats, Snort community rules.

Tool 18: Splunk / ELK / Wazuh

Category: SIEM

Overview: Log aggregation and correlation platforms for security monitoring.

Usage / Examples:

```
Ingest logs, build searches, dashboards, alerts
```

Q: What is a SIEM?
A: A system that centralises logs, correlates events and raises alerts.

Q: Open-source choice?
A: Wazuh with Elastic or OpenSearch.

Q: Key skill?
A: Writing effective queries and detection rules.

Tool 19: Terraform

Category: Cloud IaC

Overview: Declarative infrastructure-as-code tool for multi-cloud provisioning.

Usage / Examples:

```
terraform init | plan | apply | destroy
```

Q: What is state?
A: A file mapping config to real resources; store remotely with locking.

Q: plan vs apply?
A: plan previews changes; apply executes them.

Q: Modules?
A: Reusable packages of resources.

Tool 20: Ansible

Category: Automation

Overview: Agentless configuration management using YAML playbooks over SSH.

Usage / Examples:

```
ansible all -m ping | ansible-playbook site.yml
```

Q: Idempotent?
A: Running twice leaves the same end state.

Q: Inventory?
A: A list of hosts and groups.

Q: Network use?
A: Modules for Cisco, Juniper, Arista, Fortinet.

Tool 21: Docker

Category: Containers

Overview: Platform for building and running containers.

Usage / Examples:

```
docker build -t app . | docker run -d -p 80:80 app | docker ps
```

Q: Image vs container?
A: An image is a template; a container is a running instance.

Q: Security tip?
A: Run as non-root, scan images, pin versions.

Q: Compose?
A: Defines multi-container apps in YAML.

Tool 22: Kubernetes (kubectl)

Category: Containers

Overview: Container orchestration platform.

Usage / Examples:

```
kubectl get pods | kubectl apply -f app.yaml | kubectl logs pod
```

Q: Pod?
A: Smallest deployable unit with one or more containers.

Q: Service types?
A: ClusterIP, NodePort, LoadBalancer.

Q: RBAC?
A: Controls who can do what in the cluster.

Tool 23: AWS CLI / Azure CLI / gcloud

Category: Cloud CLI

Overview: Command-line interfaces to manage cloud resources.

Usage / Examples:

```
aws s3 ls | az vm list | gcloud compute instances list
```

Q: Authentication?
A: Use SSO or short-lived credentials rather than long-lived keys.

Q: Profiles?
A: Named credential sets, e.g., aws --profile dev.

Q: Scripting?
A: Combine with jq and shell for automation.

Tool 24: Prowler / ScoutSuite

Category: Cloud Security

Overview: Open-source cloud security posture auditing tools.

Usage / Examples:

```
prowler aws | scout aws
```

Q: What do they check?
A: Misconfigurations against CIS benchmarks and best practice.

Q: Permissions needed?
A: Read-only audit roles.

Q: Output?
A: HTML, JSON, CSV reports.

Tool 25: BloodHound

Category: Active Directory

Overview: Graph-based tool that maps AD privilege relationships; used by both testers and defenders.

Usage / Examples:

```
SharpHound collector -> import into BloodHound -> run path queries
```

Q: Why valuable to defenders?
A: Reveals risky attack paths so you can remove them.

Q: Data source?
A: LDAP and session information from AD.

Q: Best practice?
A: Run only with authorisation.

Tool 26: Maltego / theHarvester / Shodan

Category: OSINT

Overview: Tools for gathering public information about organisations and exposed assets.

Usage / Examples:

```
theHarvester -d example.com -b all | Shodan search: org:"Example"
```

Q: Passive?
A: Mostly yes; they use public data sources.

Q: Privacy?
A: Respect laws and scope.

Q: Use?
A: Map your own attack surface.

Tool 27: Ghidra / IDA / Volatility

Category: Reverse Engineering and Forensics

Overview: Tools for disassembling binaries and analysing memory dumps.

Usage / Examples:

```
Ghidra: import binary, auto-analyse | volatility -f mem.raw windows.pslist
```

Q: Use?
A: Malware analysis and incident response.

Q: Ghidra?
A: NSA's free reverse-engineering suite.

Q: Volatility?
A: Extracts processes, connections, and artefacts from RAM.

Tool 28: Git and GitHub Actions

Category: DevOps

Overview: Version control and CI automation.

Usage / Examples:

```
git commit | git push | workflow YAML in .github/workflows
```

Q: Why for network engineers?

A: Track config changes and enable review.

Q: Secrets?

A: Store in encrypted secrets, never in the repo.

Q: Branching?

A: Feature branches and pull requests.

F. GENERAL INTERVIEW AND EXAM Q&A

Q1: What is the difference between a vulnerability, a threat and a risk?
A: A vulnerability is a weakness, a threat is something that can exploit it, and risk is the likelihood and impact of that exploitation.

Q2: What is penetration testing vs vulnerability assessment?
A: An assessment identifies and ranks weaknesses; a penetration test attempts controlled exploitation to demonstrate real impact.

Q3: What are black, grey and white box tests?
A: Black box: no prior knowledge. Grey: partial knowledge. White: full knowledge including source and architecture.

Q4: What is the OSI model?
A: A 7-layer reference: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q5: TCP vs UDP?
A: TCP is connection-oriented and reliable; UDP is connectionless and faster with no delivery guarantee.

Q6: What is a three-way handshake?
A: SYN, SYN-ACK, ACK to establish a TCP connection.

Q7: What is NAT?
A: Translating private addresses to public ones so multiple hosts share an IP.

Q8: What is a VPN?
A: An encrypted tunnel across an untrusted network.

Q9: What is zero trust?
A: Never trust by default; verify every user, device and request.

Q10: What is the shared responsibility model?
A: The provider secures the cloud; the customer secures what they put in the cloud.

Q11: What is IAM?
A: Identity and Access Management: who can access which resource and what they can do.

Q12: What is least privilege?
A: Granting only the access needed to perform a job.

Q13: What is SQL injection?
A: Untrusted input altering database queries; prevented with parameterised queries.

Q14: What is XSS?
A: Injecting scripts into pages viewed by others; prevented with output encoding and CSP.

Q15: What is MFA?
A: Requiring two or more factors: know, have, are.

Q16: What is a rules-of-engagement document?
A: It defines scope, timing, permitted techniques and contacts for a test.

Q17: Which certifications should I plan?
A: Network+, CCNA, Security+, AWS/Azure associate, CEH, PenTest+, then OSCP/CCNP/CKA.

Q18: How do I practise legally?
A: Use your own lab, TryHackMe, Hack The Box, PortSwigger Academy, OWASP Juice Shop, and bug bounty programs within scope.

G. SUGGESTED LEARNING ROADMAP

Months 1-3	Networking fundamentals, IP and subnetting, Linux basics, Network+
Months 4-6	CCNA, switching and routing labs, firewalls, cloud fundamentals
Months 7-9	Security+, AWS/Azure associate, Docker, scripting with Python
Months 10-12	CEH or PenTest+, web app hacking labs, Kubernetes, Terraform
Year 2	OSCP, AD pentesting, cloud security, red team, specialise